AOU
الجامعة العربية المفتوحة
Arab Open University
Faculty of Computer Studies

# TM352
# Web, Mobile and Cloud Technologies

## Final Examination
## Summer Semester 2023/2024

## Date: 12/June/2025

| | |
|---|---|
| **Number of Exam Pages:** 05 (including this cover sheet) | **Time Allowed: 3 Hrs** |

## Instructions:

- Total Marks: 100
- This exam consists of 3 parts.
- **ALL questions** must be answered in the External Answer booklet.
- Be sure you write your **name and ID** on the External Answer booklet.
- **Calculators** are not allowed.

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## PART 1 (20 Marks)

Mark the following statements True(T) or False(F). Copy the answers in your answer booklet [2 Marks each]

| Statement | T | F |
|---|---|---|
| 1. Asymmetric key cryptography is used to achieve digital signature | ◯ | ◯ |
| 2. In Message Hashing technique, the hash function is chosen so that any minor change to the original message will not result in a different hash value | ◯ | ◯ |
| 3. {"name":"123"} is a valid Java Script Object | ◯ | ◯ |
| 4. In SaaS, the tenant does not manage or control the underlying cloud infrastructure but has control over operating systems, storage, and deployed applications and possibly limited control of select networking components. | ◯ | ◯ |
| 5. In OpenStack cloud architecture the images block represents the disk storage used for VM images and it is not managed by one of the services running on the controller node. | ◯ | ◯ |
| 6. AWS architecture, 'Amazon S3' is a DNS service by Amazon | ◯ | ◯ |
| 7. By Flavors, most cloud providers limited number of guest configuration to maximize the use of single host. | ◯ | ◯ |
| 8. In a SOAP web service the request is transported as URI | ◯ | ◯ |
| 9. In a responsive website when narrowing your browser window, you'll notice that there are certain widths at which the display contents change, these widths are called break points. | ◯ | ◯ |
| 10. Caesar cipher is an example of stream cipher | ◯ | ◯ |



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## PART 2 (30 Marks)

Answer the following questions.

1. Complete the missing xml elements given the below SOAP Request and Response messages. [6 marks]

*Request*

```
<?xml version="1.0" encoding="UTF-8"?><S:Envelope
xmlns:S="http://schemas.xmlsoap.org/soap/envelope/" xmlns:SOAP-
ENV="http://schemas.xmlsoap.org/soap/envelope/">
    <SOAP-ENV:Header/>
    <S:Body>
        <ns2:areaOfCircle xmlns:ns2="http://mta/">
            <radius>2.0</radius>
        </ns2:perimeter>
    </S:Body>
</S:Envelope>
```

*SOAP Response*

```
<?xml version="1.0" encoding="UTF-8"?><S:Envelope
xmlns:S="http://schemas.xmlsoap.org/soap/envelope/" xmlns:SOAP-
ENV="http://schemas.xmlsoap.org/soap/envelope/">
    <SOAP-ENV:_____/>
    <S:Body>
        <ns2:_____ xmlns:ns2="http://mta/">
            <_____>
        </ns2:_____>
    </S:Body>
</S:Envelope>
```

Table 3.1. SOAP Req and Res messages.

2. Explain the Diffie-Hellman Key Exchange process. What are the key steps involved, and why is it considered secure even if an attacker intercepts the public values? [10 Marks]

1. The NIST (National Institute of Standards and Technology) document defines three cloud service models, which are Software as a Service(SaaS), Platform as a Service(PaaS) and Infrastructure as a Service(IaaS). Discuss these three services in details. Your discussion must include what consumer can do and cannot do while using these three service models. [14 Marks]



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## PART 3 (50 Marks)

Read the scenario and solve the given four tasks.

**Scenario:**

Many expatriates in Kingdom of Saudi Arabia send money to their home country by using money transfer apps. One of such app called Money2Home uses web services to convert the currency from one form to another such as US Dollar (USD) to Saudi Riyal (SAR). The currency from SAR to other forms is stored in a JSON file named SAR2Any. The following formula is used to calculate the total amount in the target currency give the amount in SAR.

targetAmount = sourceAmount * rate equation – 1

Write Web service that accept the sourceAmount and the currency type and return the targetAmount in HTML format.

```
"SAR2Any" : [ { "type" : "INR", "rate" : "21.78"},
              { "type" : "PR", "rate" : "72.78"},
              { "type" : "BTK", "rate" : "25.38"},
              ..........}
            ]
```

SAR2Any.JSON template

**Tasks:**

1. Implement the service as RESTful service with the required annotations and read the type of currency and its rate from the JSON file called "SAR2Any.json" using the classes Json, JsonObject, and JsonArray. The file contains an array of currencies with each array element contains type of currency and its rate. [25 Marks]
2. Design the HTML page and write the HTML code that would consume or call the service. [15 Marks]
3. Specify the parameter type you have used in your implementation and the HTTP method used to call the service. [4 Marks]
4. In case that the web application should be available to mobile users, compare the development of the application as web, native, or hybrid application regarding the use of mobile hardware capabilities. Suggest the hardware capabilities that could be used for this application. [6 Marks]

**Some Json class methods that could be used:**

- static JsonArrayBuilder createArrayBuilder()//Creates a JSON array builder
- static JsonObjectBuilder createObjectBuilder()//Creates a JSON object builder
- static JsonReader createReader(InputStream in)//Creates a JSON reader from a byte stream.
- static JsonReader createReader(Reader reader) //Creates a JSON reader from a character stream.
- static JsonWriter createWriter(OutputStream out) //Creates a JSON writer to write a JSON object or array structure to the specified byte stream.
- static JsonWriter createWriter(Writer writer) //Creates a JSON writer to write a



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JSON object or array structure to the specified character stream.

**Some JsonObject methods that could be used:**

- boolean getBoolean(String name) //Returns the boolean value of the associated mapping for the specified name.
- int getInt(String name)//returns int value to which the specified name is mapped.
- JsonArray getJsonArray(String name)//Returns the array value to which the specified name is mapped.
- JsonObject getJsonObject(String name)//Returns the object value to which the specified name is mapped.
- String getString(String name)//Returns the string value to which the specified name is mapped

**Some JsonArray methods that could be used**

- int getInt(int index) //A convenience method for getJsonNumber(index).intValue().
- JsonArray getJsonArray(int index) //Returns the array value at the specified position in this array.
- JsonObject getJsonObject(int index) //Returns the object value at the specified position in this array.
- String getString(int index) //Returns the String value at the specified position in this array.

**Some JsonReader methods that could be used**

- void close() //Closes this reader and frees any resources associated with the reader.
- JsonArray readArray() //Returns a JSON array that is represented in the input source.
- JsonObject readObject() //Returns a JSON object that is represented in the input source.

**Some JsonWriter methods that could be used**

- void close() //Closes this JSON writer and frees any resources associated with the writer.
- void writeArray(JsonArray array) //Writes the specified JSON array to the output source.
- void writeObject(JsonObject object) //Writes the specified JSON object to the output source.

**End of questions**



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